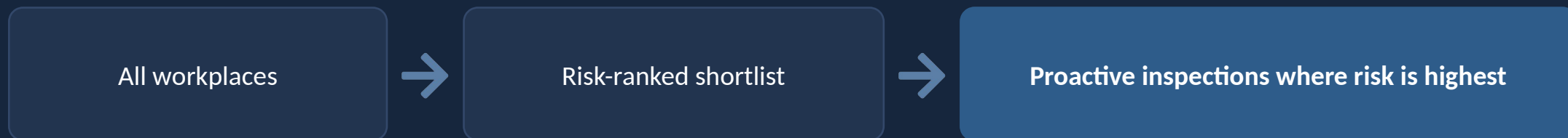


Targeting Inspections Where Risk Is Highest

Predictive Risk Identification — a workplace-level risk-scoring engine for proactive inspection planning



- Inspections today are largely reactive — after a complaint, injury, or fatality — or broad sector campaigns
- Inspector capacity is finite; most workplaces are never visited in a given year
- The prize: more serious problems caught — and injuries prevented — per inspector-day

THE SOLUTION AT A GLANCE

A Risk-Scoring Engine for Inspection Planning

Combine enforcement history with inspection-report text to produce a ranked, explainable shortlist.



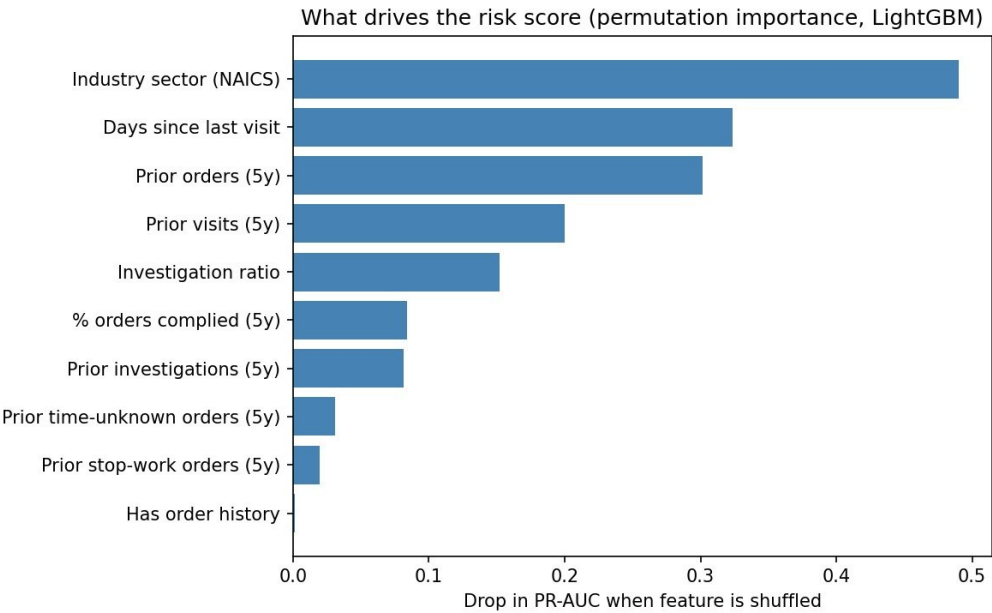
Designed for the Ministry's existing stack

Azure · Databricks · Power BI

What the Data Says Drives Risk

 **Target (precise):** likelihood of a **serious enforcement action** (stop-work / time-unknown order) on inspection — **not** individual accidents.

Top structured drivers



Prior orders & investigations, recency, compliance rate, and industry sector (construction & mining highest).

Text adds what structured fields miss

BEFORE — de-identified report excerpt

A worker was critically injured when steel tubes fell during unloading. The worker was 21, employed two weeks, working without direct supervision. The employer had no documented training program for the task.

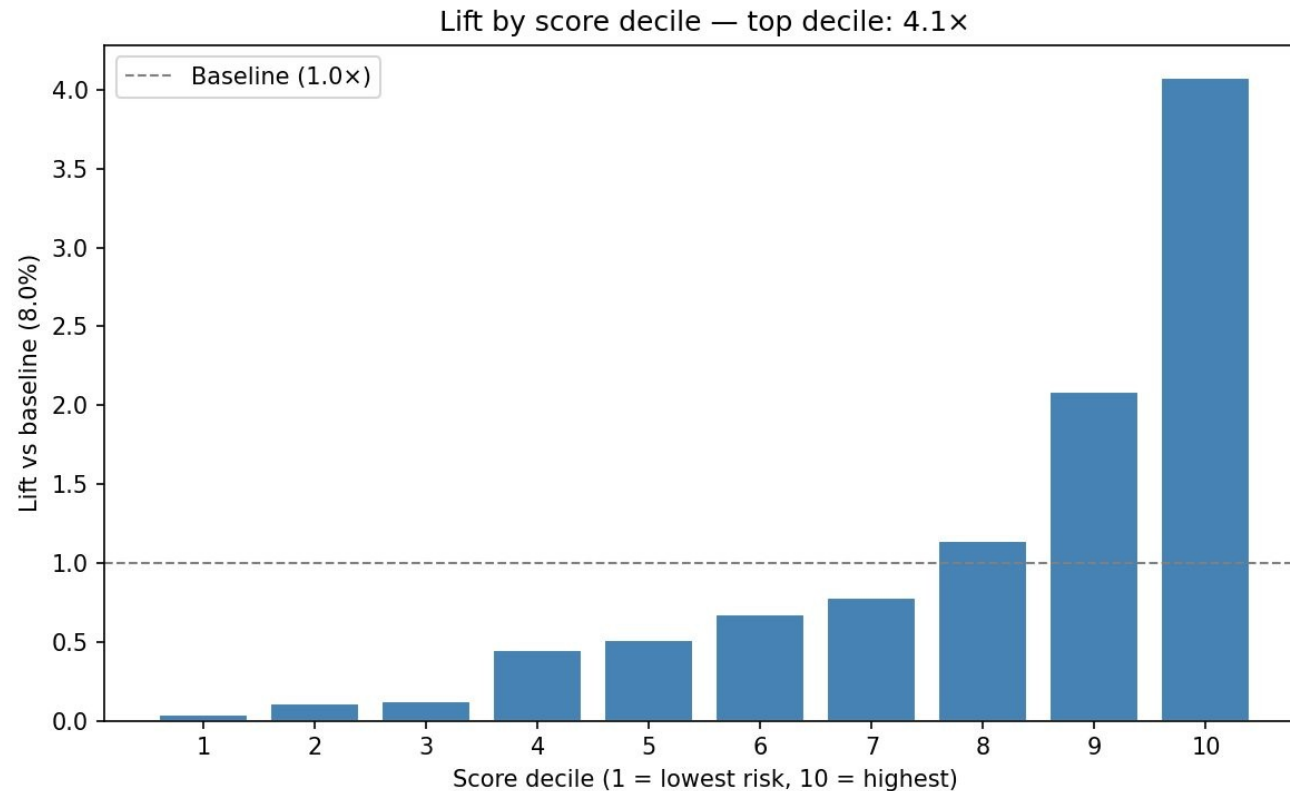


AFTER — LLM extraction (schema-constrained JSON)

```
"hazard_type": "caught-in / crush"
"injury_severity": "critical"
"vulnerable_worker_flag": true
"supervision_failure_flag": true
"systemic_failure_flag": true
"order_sections": ["OHSA 25(2)(a)", ...]
```

Narratives become features and dashboard filters: supervision failure, young/new worker, systemic cause.

Validated the Way It Would Be Used



Trained on 2017–2022, tested on 2023; cross-validated, not in-sample. Monotonic lift across deciles.

~4x

lift in the top decile — serious problems found per inspection vs untargeted visits

32% vs 8%

serious-order rate: top-scored workplaces vs the 8.0% baseline

- ✓ **Robustness:** gradient boosting didn't beat the simple model out-of-fold → signal is real; we keep the fully explainable model with no accuracy penalty.
- ✓ **Business translation:** same harm prevented with fewer visits — or more harm prevented with the same budget.

Built for a Government Environment

Private, fair, explainable, human-in-the-loop, and deployable on our stack.



Privacy (FIPPA / PHIPPA)

De-identify reports before processing; LLM in-tenant on Azure — no data to public APIs.



Fairness

Audit scores for disparate impact by region, sector, and firm size.



Avoid feedback bias

Reserve random / exploratory inspections so the model keeps learning the whole population.



Human-in-the-loop

The model ranks; inspectors decide. Reason codes make every score reviewable.

Roadmap

1

Pilot one sector / region

2

Validate

3

Scale on Databricks

4

Monitor drift & fairness

A1 · Target Definition & Selection Bias

Target variable

Serious order in 2023 = Stop Use / Stop Work Order OR Time Unknown Order. These signal immediate danger.

Evaluation cohort

12,234 workplaces with enforcement history 2017–2022 AND at least one 2023 visit. Baseline serious-order rate: 8.0%.

Why labels are MNAR (missing not at random)

Outcomes only exist where we chose to inspect. Scores for never-visited sites are extrapolations. Mitigation: reserve exploratory inspections; validate against near-random campaign visits.

A2 · Model Metrics — Logistic vs LightGBM

Metric	Logistic (kept)	LightGBM
Lift @ top 10% (in-sample)	4.08×	—
Lift @ top 10% (cross-validated)	4.00×	3.73×
Top-decile serious-order rate	~32%	—
PR-AUC (CV)	0.31	—
ROC-AUC (CV)	0.80	—

Boosting did not beat logistic out-of-fold → keep the explainable model. Numbers are cross-validated on public open data; directional, not a production guarantee.

A3 · Feature Dictionary (top drivers)

Feature	Definition
Industry sector (NAICS)	Sector code; construction & mining carry highest baseline risk
Days since last visit	Recency of last inspection contact
Prior orders (5y)	Count of enforcement orders issued in prior 5 years
Prior visits (5y)	Count of inspection visits in prior 5 years
Investigation ratio	Share of visits that escalated to investigation
% orders complied (5y)	Compliance behaviour on past orders
Prior investigations (5y)	Count of investigations in prior 5 years
Prior stop-work orders (5y)	History of the most serious order type

A4 · LLM Extraction — Schema & FIPPA-Safe Prompt

Controls: de-identify before processing · data minimization · in-tenant Azure OpenAI · access controls · schema-constrained JSON only. The LLM structures text — it never assigns enforcement.

```
SYSTEM: Extract safety attributes from the de-identified inspection
report. Return ONLY valid JSON matching the schema. No personal data.
```

```
{
  "hazard_type": "<caught-in|struck-by|fall|...>",
  "injury_severity": "<none|minor|serious|critical|fatal>",
  "vulnerable_worker_flag": <bool>,
  "supervision_failure_flag": <bool>,
  "systemic_failure_flag": <bool>,
  "order_sections": ["<OHS/Reg citation>", ...]
}
```